Introduction to Problem Solving

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Computer Science is a science of abstraction creating the right model for a problem and devising the appropriate mechanizable technique to solve it.

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Steps for Problem Solving

Suppose while driving a vehicle starts making a strange noise. We might not know how to solve the problem right away. First we need to identify from where the noise is coming. In case the problem cannot be solved by us, then we need to take the vehicle to a mechanic. The mechanic will analyze the problem to identify the source of the noise, make a plan about the work to be done and finally repair the vehicle in order to remove the noise. From the example it is explicit that finding solution to a problem might consist of multiple steps.

Key steps to solve the problem

1. Analyzing the problem.
2. Developing an algorithm.
3. Coding.
4. Testing and debugging.

1. Analyzing the problem

It is important to clearly understand a problem before we begin to find the solution for it. If we are not clear as to what is to be solved and we may end up developing a program which may not be solved our purpose. Thus we need to read and analyze the problem statement carefully in order to list the principal components of the problem and decide the core functionalities that our solution should have. By analyzing a problem, we would be able to figure out what are the inputs that our program should accept and the output that it should produce.

2. Developing an algorithm

It is essential to device a solution before writing a program code for a given problem. The solution is represented in natural language and is called an algorithm. We can imagine an algorithm like a very well written recipe for a dish, with clearly defined steps that if followed one will end up preparing the dish.

In other words, "a set of exact steps which, when followed, solve the problem or accomplish the required task" is called Algorithm.

3. Coding

After finalizing the algorithm, we need to convert the algorithm into the format which can be understood by the computer to generate the desired solution. Different high level programming language can be used for writing a program. It is equally important to record the details of the coding procedures followed and document the solution. This is helpful when revisiting the program at a later stage.

4. Testing and debugging

The program created should be tested on various parameters. The program should meet the requirement of the user. It must respond within the expected time. It should generate correct output for all possible inputs. In the presence of syntactical errors, no output will be obtained. In case the outputs generated is incorrect, then the program should be checked for logical errors, if any.

Software industries follows standardized testing methods like unit or component testing, integration testing, system testing and acceptance testing while developing complex application. This is to ensure that software meets all the business and technical requirements and works as expected. Debug or rectified and the program is again tested. This continues till all the errors are removed from the program.

Once the software application has been developed and tested. And deliver to the user. Still problems in terms of functioning can come up and needs to be resolved from time to time. The maintenance of the solution thus involves fixing the problem faced by the user. Answering the queries of the user and even serving the request for addition or modification of features.

Algorithm

In our day to day life, we perform activities by following certain sequence of steps. Example of activities include getting ready for school, making breakfast, riding a bicycle, wearing a tie, solving a puzzle and so on. To complete each activity, we follow a sequence of steps. Suppose following are the steps required for an activity 'Riding a bicycle':

i. Remove the bicycle from the stand.
ii. Sit on the seat of the bicycle.
iii. Start peddling.
iv. Use brakes whenever needed and
v. Stop on reaching the destination.

We need to find GCD(Greatest Common Divisor) or HCF(Highest Common Factor):
GCD: Is the largest number that divides both the given numbers.

i. Store two numbers in n1 and n2
ii. While n2 > 0: True
Fake → a) R = n1 mod n2 --->Remark: mod will give remainder
→ b) n1 = n2
→ c) n2 = R
iii. Print "GCD = ",n1
iv. End

n1 = ~~27~~ ~~72~~ ~~27~~ ~~18~~ 9
n2 = ~~72~~ ~~27~~ ~~18~~ 9 0
R = ~~27~~ ~~18~~ 9 0

72 | 27 | 0
−00
→ 27 | 72 | 2
54
18 | 27 | 1
18

iii. Print "GCD = ",n1
iv. End

n2 = ~~72~~ ~~27~~ ~~18~~ ~~9~~ ~~0~~
R = ~~27~~ ~~18~~ ~~9~~ 0

54
18 | 27 | 1
18
9 | 18 | 2
18
0

GCD or HCF

The origin of the term algorithm is traced to Persian astronomer and mathematician Abu Abdul Law Mohammad Even Musa Al Khwarizmi.(c850AD) as the latin translation of Al-Khwarizmi was called 'Algorithm'.

Why do we need an Algorithm?
So a programmer writes a program to instruct the computer to do certain task as desired. The computer then follows the steps written in the program code. Therefore, the programmer first prepares a road map of the program to be written before actually writing the code. Without a road map the programmer may not be able to clearly visualize the instructions to be written, and may end up developing a program which may not work as expected.
Such a road map is nothing but the algorithm which is the building block of a computer program. For example, searching using a search engine, sending a message, finding a word in a document, booking a taxi through an app, performing online banking, playing computer games are all based on algorithms.
Writing an algorithm is mostly considered as the first step to programming. Once we have an algorithm to solve a problem, we can write the computer program forgiven instructions to the computer in high level language. If the algorithm is correct, computer will run the program correctly every time. So the purpose of using an algorithm is to increase the reliability, accuracy and efficiency of obtaining solutions.

Characteristics of a good algorithm

- **Precision-** The steps are precisely stated or defined.
- **Uniqueness-** Results of each step are uniquely defined and only dependent on the input and the results of the preceding steps.
- **Finiteness-** The algorithm always stops after a finite number of steps.
- **Input-** The algorithm receives some input.
- **Output-** The algorithm produces some output.

While writing an algorithm, it is required to clearly identify the following:

- The input to be taken from the user.
- Processing or computation to be performed to get the desired result.
- The output desired by the user.